

UNIT-II (Chapter-I)
**SCIENTIFIC AND
TECHNOLOGICAL
DEVELOPMENTS
IN MEDIEVAL INDIA**

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OBJECTIVES

This chapter investigate the growth of sciences in India during the period from 8th to 18th century A.D.

After reading this lesson you will be able to:

1. discuss the educational practices that emerged during the medieval period;
2. trace the developments in science & technology in Medieval India;
3. assess the influence of Arabian and Persian science in India in the period under discussion and
4. list some well known scholars in the field of science & technology and their works during this period.

Introduction

Muslim rule started in India with the onset of 13th century A.D. and lasted for around five hundred year and concluded with the death of Emperor Aurangzeb in 1707.

- 1.**During this period considerable work was done in various sphere of science and technology. Sciences such as a) mathematics, b)medicine, c)astrology, d)astronomy flourished.
- 2.**It is also true that in Medieval India, science was not patronized as a state policy by Kings or the Raja's.
- 3.** It is also unfortunate that science and technology was not pursued rigorously as it was being developed in Europe.
- 4.**Huge monuments were constructed, observatory were built, texts were translated but the volume was too low in comparison to the west.
- 5.** In spite of all these hurdles, sciences flourished in a scanty manner.
- 6.**This chapter endeavors to throw light on the influence of Arabian world and subsequent growth of various aspects of science during medieval phase of Indian history.

Sciences in Medieval Age

- Muslim rule started in India with the onset of 13th century A.D. and lasted for around five hundred year and concluded with the death of Emperor Aurangzeb in 1707.
- By this time, the traditional indigenous classical learning had already received a setback. The pattern of education as prevalent in Arab countries was gradually adopted during this period.
- Maktabas and Madrasas came into existence. Arithmetic, Mensuration, Geometry, Astronomy, Accountancy, Public Administration and Agriculture were included in the courses of studies for primary education.
- Large workshops called karkhanas were maintained to supply provision, stores and equipments to royal household and government departments.
- The karkhanas not only worked as manufacturing agencies, but also served as centres for technical and vocational training to young people.
- The karkhanas trained and turned out artisans and craftspersons in different branches, who later on set up their own independent karkhanas.

Mathematics

Several works in the field of Mathematics were produced during this period.

- Narayana Pandit, son of Narsimha Daivajna was well known for his works in Mathematics-Ganitakaumudi and Bijaganitavatamsa.
- Gangadhara, in Gujarat, wrote Lilavati Karamdipika, Suddhantadipika , and Lilavati Vyakhya.
- These were famous treatises which gave rules for trigonometrical terms like sine, cosine tangent and cotangent.
- Nilakantha Somasutvan produced Tantrasamgraha, which also contains rules of trigonometrical functions.

Mathematics contd...

- Ganesa Daivajna produced Buddhivilasini-a commentary on lilavati-containing a number of illustrations.
- Krishna of the Valhalla family brought out Navankura on the Bijaganit of Bhaskara-II and elaboration of the rules of indeterminate equations of the first and second orders.
- Nilakantha Jyotirvida 46 compiled Tajik, introducing a large number of Persian technical terms. Faizi, at the behest of Akbar, translated Bhaskara's Bijaganit.
- Akbar ordered to make Mathematics as a subject of study, among others in the education system . Naisiru'd -din-at -tusi, was another scholar of Mathematics.

Chemistry

- Do you know that in the medieval period, use of paper had begun?
- An important application of Chemistry was in the production of paper.
- Kashmir, Sialkot, Zafarabad, Patna, Murshidabad, Ahmedabad, Aurangabad and Mysore became well known centres of paper production.
- The paper making technique was more or less the same throughout the country differing only in preparation of the pulp from different raw materials.
- The Mughals knew the technique of production of gunpowder and its use in gunnery, another application of Chemistry.
- The Indian crafts persons learnt the technique in evolved suitable explosive composition. The work Sukraniti attributed to Sukracarya contains a description of how gunpowder can be prepared using saltpeter, sulphur and charcoal in different ratios for use in different types of guns..
- The principal type of fireworks included those which pierce through air, produce sparks of fire, blaze with various colours and end with explosion.
- The work Ain-I-akbari speaks of the regulation of the Perfume office of Akbar. The attar (perfume) of roses was a popular perfume, which is supposed to have been discovered by Nurjehan.

Astronomy

- Astronomy was another field that flourished during this period.
- In astronomy, a number of commentaries dealing with the already established astronomical notions appeared.
- Mehendra Suri, a court astronomer of Emperor Firoz Shah, developed an astronomical instrument ‘Yantraja’.
- Paramesvara and Mahabhaskariya, both in Kerala, were famous families of astronomers and almanac-makers.
- Nilakantha Somasutvan produced commentary of Aryabhatiyaa. Kamalakar studied the Islamic astronomical ideas. He was an authority on Islamic knowledge.
- Maharaja Sawai Jai Singh-II of Jaipur was a patron of Astronomy.
- He set up the five astronomical observatories in Delhi, Ujjain, Varansasi, Mathura and Jaipur.

Medicine

- The Ayurveda system of medicine did not progress as vigorously as it did in the ancient period because of lack of royal patronage.
- However, some important treatises on Ayurveda like the Sarangdhara Samhita and Chikitsasamgraha by Vangasena, the Yagaratbajara and the Bhavaprakasa of Bhavamisra were compiled.
- The Sarangdhara Samhita, written in the thirteenth century, includes use of opium in its material medica and urine examination for diagnostic purpose.
- The drugs mentioned include metallic preparation of the rasachikitsa system and even imported drugs.
- The Rasachikitsa system, dealt principally with a host of mineral medicines, both mercurial and non mercurial.
- The Siddha system mostly prevalent in Tamil Nadu was attributed to the reputed Siddhas, who were supposed to have evolved many life-prolonging compositions, rich in mineral medicines.

Medicine contd...

- The Unani Tibb system of medicine flourished in India during the medieval period.
- Ali- in Rabban summarized the whole system of Greek medicine as well as the Indian medical knowledge in the book, Firdausu-Hikmat.
- The Unani medicine system came to India along with the Muslims by about the eleventh century and soon found patronage for its growth.
- Hakim Diya Muhammad compiled a book, Majiny-e Diyae, incorporating the Arabic, Persian and Ayurvedic medical knowledge. Firoz Shah Tughalaq wrote a book, Tibbe Firozshahi.
- The Tibbi Aurangzebi, dedicated to Aurangzeb, is based on Ayurvedic sources.
- The Musalajati-Darshikohi of Nuruddin Muhammad, dedicated to Darashikoh, deals with Greek medicine and contains, at the end, almost the whole of Ayurvedic material medica.

Agriculture

- In the medieval period, the pattern of agricultural practices was more or less the same as that in early India.
- Some important changes occurred in the introduction of new crops, trees as well as horticultural plants by foreign traders.
- The principal crops were wheat, rice, barley, millets, pulses, oilseeds, cotton, sugar-cane and indigo.
- The Western Ghats continued to yield black pepper of good quality and Kashmir maintained its tradition for saffron and fruits.
- Ginger and cinnamon from Tamil Nadu, cardamom, sandalwood and coconut from Kerala, were becoming increasingly popular.
- Tobacco, chillies, potato, guava, custard apple, cashew and pineapple were the important plants which were introduced to India during the sixteenth and seventeenth centuries.
- It was during this period that the production of opium from poppy plants began in Malwa and Bihar regions.
- Improved horticultural methods were adopted with great success.
- The systematic mango grafting was introduced by the Jesuits of Goa in the middle of sixteenth century.
- Imperial Mughal Gardens were suitable areas where extensive cultivation of fruit trees came up.

Agriculture contd...

- For irrigation, wells, tanks, canals, rahat, charas and dhenkli charas (a sort of a bucket made of leather used to lift water with the help of yoked oxen) were used.
- Persian wheel was used in the Agra region.
- In the medieval period, agriculture was placed on a solid foundation by the State by introducing a system of land measurement and land classification, beneficial both to the rulers as well as the tillers.

Influence of India on Arabian World The scientific cooperation between India and the Arabs dates back to the time of Abbasid Caliphate of Baghdad when a number of books on astronomy, mathematics, and medicine were translated from Sanskrit into Arabic. The cooperation began with the time of Caliph al-Mansur (CE.753-774). Translation of Brahmasiddhanta, derived from it, foundation for computing the motions of the planets. 750 CE., book was called Al-Zij 'ala Sini al-'Arab, or Sindhand al-Kabir. With the help of Hindu Pundits, Al-Fazari, translated Brahmagupta's other book Khandakhadyaka and gave it the Arabic name of Arkand. It was on this occasion that the Arabs first became acquainted with the Hindu system of astronomy. They learned astronomy from Brahmagupta (7th Cent.AD) earlier than Alexandrian scientist Ptolemy. What we call today Arabic numerals, were in fact Indian numbers. Arab scientists in Iraq, especially Muhammad ibn Musa al-Khawrizmi (CE 9th Cent) used the new numbers to develop algebra. Some mathematical and astronomical terms were borrowed from Sanskrit. Ethical writings of Chanakya (Shanaq) and works on logic and magic were translated as catalogued by Ibn Nadim in his 10th century Kitab-al-Fihrist. Ibn al-Muqaffa translated Pancatantra into Arabic as Kalila wa Dimna. The fascinating story of Sindbad was partly of Indian origin. Parts of Mahabharata were rendered into Arabic by Ali Jabali, in C.1026. Translation of Susrata Samhita into Arabic. For over five hundred years Muslim & other writers continued to learn and apply.

During the Mughal rule of India, science & technology developed mainly due to the interests of Emperors and Sultans, particularly in astronomy, agriculture, engineering, architecture and medicine. A number of encyclopedias and dictionaries were penned. Initially dictionaries were needed as new ideas were being developed as a result of interaction between Sanskrit and other languages. Al-Biruni regarded the essence of Hindu religion as a form of monotheism, idol worship as ignorant passions of the people. He was the first to introduce the study of Bhagavad Gita to the Muslim world, and the first Muslim to study the Puranas and to translate Patanjali and Samkhya into Arabic.

Influence of Arabian Sciences in India

Centralised Governance, Islamic Law (Sharia), The "Zimmi" Status: The concept of *Zimmis* (protected non-Muslim subjects against Ziziya tax), **Yunani Tibb (Ionian Medicine)**,

Technological Innovations: India adopted the **spinning wheel** (*charkha*, *invented in China or India around 500 BCE*) (why was Nehru against Gandhi?) from the Islamic world by the 13th century, in a large scale because of fashion need, which revolutionized the textile industry. Other adoptions included advanced **irrigation techniques** like the Persian wheel (Noria) and improved paper-making technology.

Commerce and Urban Planning

- **New Trade Networks,**
- **Banking & Book-keeping: double-entry book-keeping and Hundis.**
- **Urban Design: Mosque Construction**
- **Sufism, Culinary, Fashion**

Curiosity of Sultans of Delhi

Sultan Muhammad bin Tughlaq (1351) was a great scholar versed in logic, Greek philosophy, mathematics, astronomy and physical sciences. He had knowledge of medicine and was skillful in dialectics. He also was an expert calligrapher. He enjoyed the society of Hindu yogis and extended his patronage to Jain divines. The Sultans of Delhi were very much interested in mechanical machines like pulleys and piers. In the book Sirat Feroz Shahi (1370) 13 such instruments were listed which were used in transporting stones and heavy building materials. He set up hospitals for free treatment of the poor and encouraged physicians in the development of Unani medicine. He ordered a work on Hindu astronomy and astrology to be translated into Persian under the name of Dalaile Firoz Shahi. Sultan Zain al-Abidin of Kashmir (1420-1470) who commissioned the translations of Mahabharata and Raja-tarangini into Kashmiri language, which was first indication of Muslim interest in the pre-Muslim Hindu history of India.

Intellectual curiosity of Emperor Akbar The Mughal Emperors (1526-1858) took a keen interest in the development of astronomy. They patronized astronomers in their royal courts. The works thus produced were mainly zijes (astronomical tables) and calendars. Many scientific works brought from outside of India. Attempts were made to write commentaries and translate these works. As a result, the intermingling of Indian mathematical tradition with Arabic & Persian did take place enriching the country. Emperor Humayun (1556) built a personal observatory near Delhi, while Jahangir and Shah Jahan were also intending to build observatories but were unable to do so. Mulla Chand, a court astronomer of Emperor Nasiruddin Humayun produced —Tashil Mulla Chandll, which was a redaction of Zije Ulugh Beg. Muslim patronage of Hindu learning reached its highest watermark in the court of Emperor Jalal al-Din Akbar (d.1605). Some of the Hindu nobles in his court wrote in Persian and Sanskrit, like Raja Manohardas and Raja Todar Mal (d.1589) who translated Bhagavata Purana into Persian. Akbar had a stupendous library composed entirely of manuscripts written and engraved by skilful penmen. The volumes in his library numbered only 24,000 but they valued at \$3,500,000. He patronized poets and learned men. He supervised the translation of Mahabhartta into Persian. In the preface to his Persian translation of Mahabharata, Abul Fazl says: —Akbar initiated a policy so that in his age the pillars of blind following were demolished and a new era of research and enquiry in religions matters commenced. In 1578 he ordered Abul Fazl to translate the New Testament into Persian. No copy of this translation is extant, but it appears he made the translation with the help of the Catholic Fathers. The translation of Ramayana was undertaken by Abdul Qadir Badauni on the express command of Akbar in 1585 and completed in 1590. The Harivamsa Purana, supplement to Mahabhartta, was translated by eminent Persian poet Mulla Sheri.

Intellectual curiosity of Emperor Akbar (contd..)

Shaikh Abu al-Faiz ibn Mubarak-pen name Faizi (1547-95) was a poet laureate of Emperor Akbar. At the suggestion of Akbar, Faizi translated Bhaskar Acarya's (1114-60) Sanskrit work on mathematics Lilavati into Persian in 1587; containing theorems of arithmetic and algebra. The translation was so popular that Ataullah Rashdi Lahori translated Bhaskar Acarya other books on algebra and measurement. Faizi, a prodigious author of 100 books, translated few mathematical problems from Latin into Persian also. The famous book covering the administration of Emperor Akbar, Aln-I-Akbari written by Abul Fazl Allami ibn Mubarak (d.1602), described West and Central Asian astronomy. Abu-al-Fa'iz's greatest literary accomplishment was the monumental Akbar-nāmah in 3 volumes. Among his many works is a Persian translation of the Bible. Authors of later generations admired his style and sought to imitate it. Zije Ulugh Beg, prepared by Sultan Ulugh Beg (1393-1449) in Samarkand was translated into Sanskrit, entitled Ulakabegijica.

Knowledge System The pattern of education as prevalent in Arab countries was gradually adopted during this period. As a result, Maktabas and Madrasas came into existence. These institutions used to receive royal patronage. Apart from the talent available locally in the country, learned men from Arabia, Persia and Central Asia were also invited to take charge of education in madrasas. Some important subjects like Arithmetic, Mensuration, Geometry, Astronomy, Accountancy, Public Administration and Agriculture were included in the courses of studies for primary education. Though special efforts were made by the ruler to carry out reforms in education, yet sciences did not make much headway. Efforts were made to seek a kind of synthesis between the Indian traditional scientific culture and the prevalent medieval approach to science in other countries. The karkhanas not only worked as manufacturing agencies, but also served as centres for technical and vocational training to young people. The karkhanas trained artisans and crafts persons, who later on set up their own independent karkhanas. The Royal workshops (Karkhanas) played an important role in the administrative system of the Sultanate. The needs of the royal household were met through Karkhanas. The Karkhanas were of two types - (i) Manufacturing place (ii) Store House. Under Feroz Tughlaq, there were as many as 36 Karkhanas. Each Karkhana was supervised by a noble who had the rank of a Malik or a Khan. It is unfortunate science and technology was not pursued rigorously as it was being developed in Europe.

Conclusion

- During the Muslim rule of India considerable work was done in mathematics, medicine, astrology, astronomy, and translations of various texts.
- Custodians of faith filled the minds of people with superficial things and did not allow enquiry into religious dogmas.
- Science was not patronized as a state policy by Kings or the Raja's. It is unfortunate science and technology was not pursued rigorously as it was being developed in Europe.
- No scientific institutions were set up, nor were students sent to Europe for higher studies. The money that was spent on constructing monumental edifices, had it been spent on creating scientific institutions, India could have become an advanced country long time ago.

Summary

- There was a considerable change in the education system. The Arabic system was introduced in a big way. Madrasas and Maktabas were established all over. The rulers tried to introduce reforms.
- Several works were written in the fields of Mathematics, Chemistry, Biology, Astronomy and Medicine.
- Most of the scientific works in this period were commentaries or expositions of the earlier treatises.
- Several important scientific works in astronomy, medicine and other sciences were rendered from Sanskrit to Persian/Arabic and from Persian/Arabic to Sanskrit.

Thank you